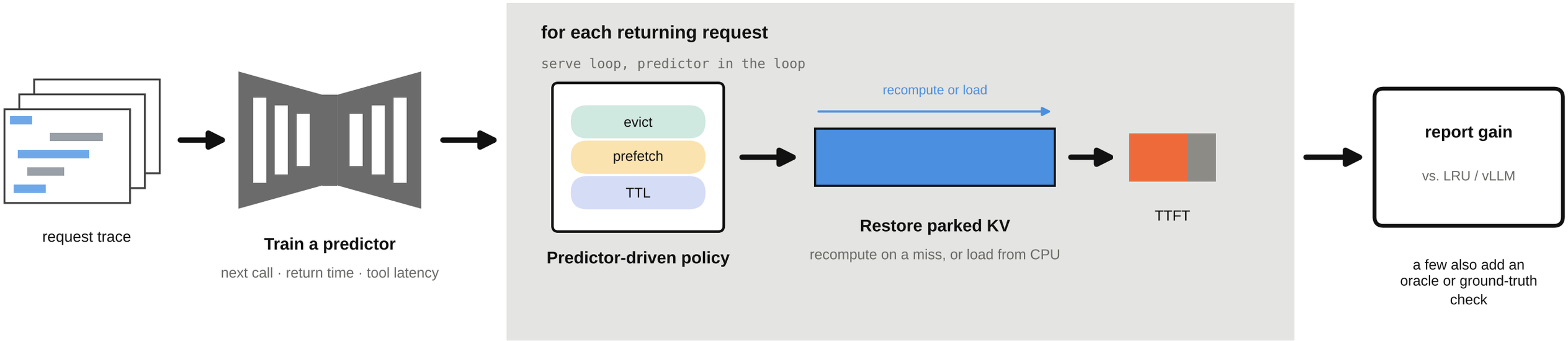
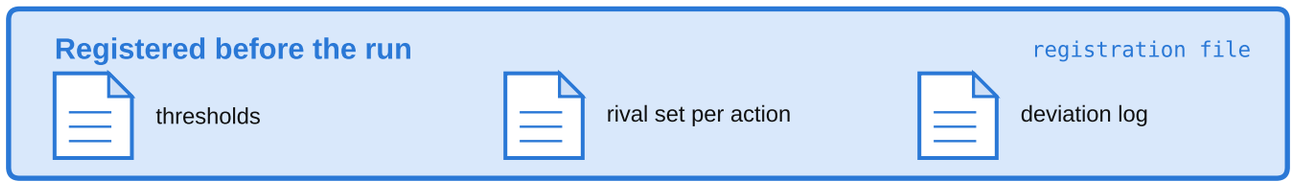
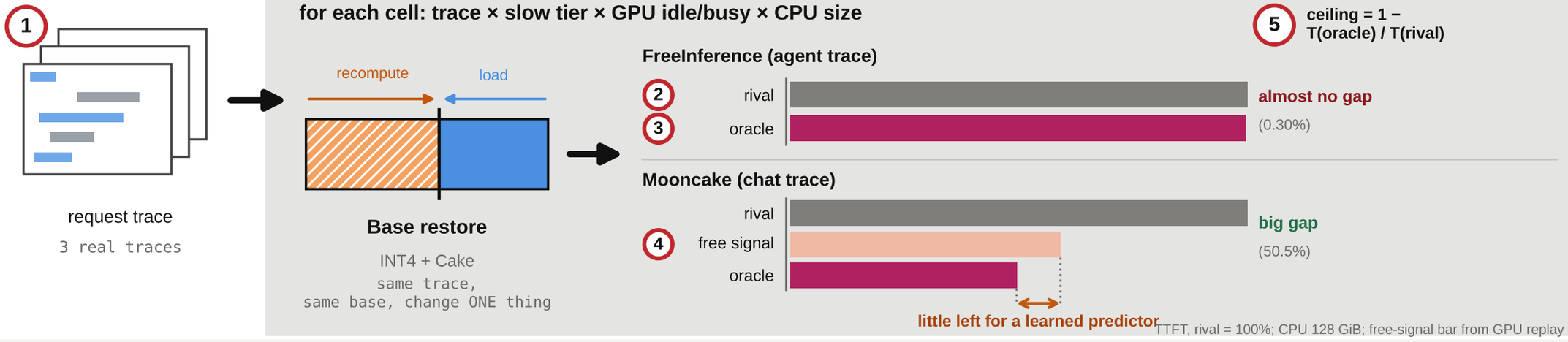
(a) Prediction-based KV management (e.g. PBKV, CacheScout, CacheWise, SAGA)



(b) Ours: measure what prediction can still add on top of the fixed base





	prediction-based work (baselines checked in the full texts)	ours
1 **Input**	a request trace or agent workflow	same trace + the best fixed restore (INT4 + Cake) as the base
2 **Compare**	against LRU, vLLM or SGLang prefix caching	against the best non-predictive rival for the same action
3 **Foresight**	a learned or pattern-based predictor (next call, tool latency)	an oracle that knows ONE future quantity at a time
4 **Free signal**	rare (e.g. tool-reported progress)	user starts typing → prefetch; gets most of the perfect-prefetch gain
5 **Measure**	gain vs. the baselines; a few add a ground-truth or oracle check	ceiling = 1 − T(oracle)/T(rival); thresholds registered beforehand
6 **Output**	a predictor and its reported gain	how much prediction can still add, per trace